

P Block Elements

Question1

Given below are two statements :

Statement I: Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II: Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the most appropriate answer from the options given below :

NEET 2025

Options:

- A. Statement I is correct but Statement II is incorrect
- B. Statement I is incorrect but Statement II is correct
- C. Both Statement I and Statement II are correct
- D. Both Statement I and Statement II are incorrect

Answer: A

Solution:

The explanation revolves around the properties and chemical behavior of group 15 elements, specifically nitrogen, arsenic, and antimony.

Statement I: This statement is correct. All elements in group 15, including nitrogen and arsenic, can form hydrides with the formula EH_3 . For instance, nitrogen forms ammonia (NH_3), and arsenic forms arsine (AsH_3).

Statement II: This statement is incorrect. Elements in group 15 are capable of forming two types of oxides: E_2O_3 and E_2O_5 . Antimony, being a group 15 element, can indeed form antimony pentoxide (Sb_2O_5).

Therefore, Statement I is correct, while Statement II is incorrect.

Question2

Match List-I with List-II :

List-I (Oxoacids of Sulphur)	List-II (Bonds)
A. Peroxodisulphuric acid	I. Two S – OH, Four S = O, One S – O – S
B. Sulphuric acid	II. Two S – OH, One S = O
C. Pyrosulphuric acid	III. Two S – OH, Four S = O, One S – O – O – S
D. Sulphurous acid	IV. Two S – OH, Two S = O

Choose the correct answer from the options given below.

[NEET 2023]

Options:

A.

A-III, B-IV, C-I, D-II

B.

A-I, B-III, C-IV, D-II

C.

A-III, B-IV, C-II, D-I

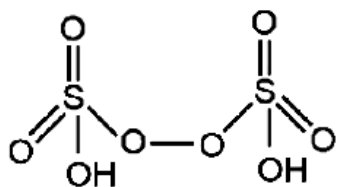
D.

A-I, B-III, C-II, D-IV

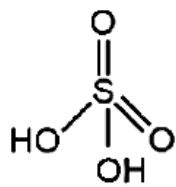
Answer: A

Solution:

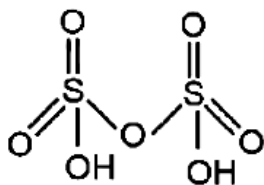
Solution:



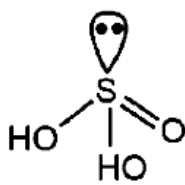
Peroxodisulphuric acid
($\text{H}_2\text{S}_2\text{O}_8$)



Sulphuric acid
(H_2SO_4)



Pyrosulphuric acid
($\text{H}_2\text{S}_2\text{O}_7$)



Sulphurous acid
(H_2SO_3)

Question3

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Helium is used to dilute oxygen in diving apparatus.

Reason R : Helium has high solubility in O_2 .

In the light of the above statements, choose the correct answer from the options given below

[NEET 2023]

Options:

A.

Both A and R are true and R is NOT the correct explanation of A

B.

A is true but R is false

C.

A is false but R is true

D.

Both A and R are true and R correct explanation of A

Answer: A

Solution:

Solution:

Helium is used as diluent for oxygen in modern diving apparatus because of its very low solubility in blood. Gases diffuse easily with each other.

Question4

Given below are two statements: one is labelled as Assertion(A) and the other is labelled as Reason (R).

Assertion (A) : ICl is more reactive than I_2 .

Reason (R): I-Cl bond is weaker than I-I bond.

In the light of the above statements, choose the most appropriate answer from the options given below:

[NEET-2022]

Options:

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A).
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

Answer: A

Solution:

Solution

In general, interhalogen compounds are more reactive than halogens (except fluorine). This is because $X-X'$ bond in interhalogens is weaker than $X-X$ bond in halogens excepts $F-F$ bond. Therefore I-Cl is more reactive than I_2 because of weaker I-Cl bond than I-I bond.

Question5

Given below are two statements Statement I The boiling points of the following hydrides of group 16 elements increases in the order - $H_2O < H_2S < H_2Se < H_2Te$

Statement II The boiling points of these hydrides increase with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below :
[NEET-2022]

Options:

- A. Both Statement I and Statement II are correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but Statement II is incorrect
- D. Statement I is incorrect but Statement II is correct

Answer: B

Solution:

Solution:

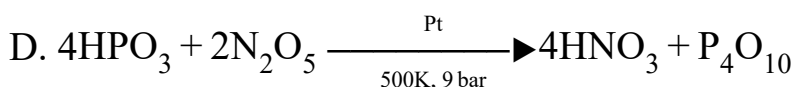
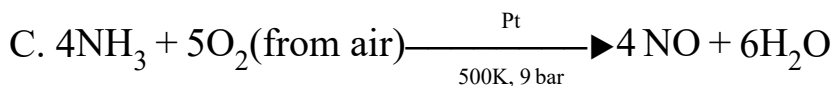
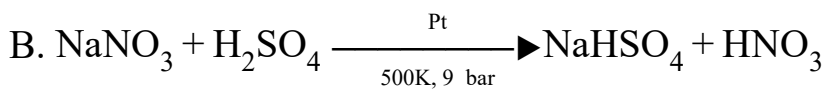
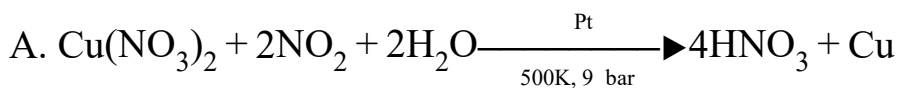
Compound	Boiling point (K)
H_2O	373
H_2S	213
H_2Se	232
H_2Te	269

- The boiling points of these hybrids not exactly increases with increase in molar mass.
- H_2O has maximum boiling point due to intermolecular hydrogen bonding.

Question6

Which of the following reactions is a part of the large scale industrial preparation of nitric acid
[NEET Re-2022]

Options:

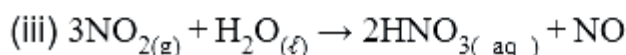
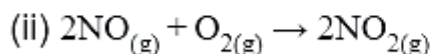
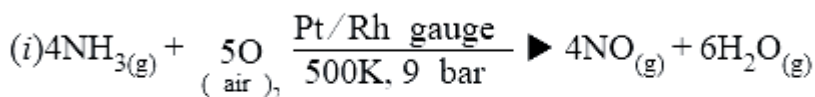


Answer: C

Solution:

Solution:

On large scale, nitric acid is prepared by Ostwald's process.



Question7

Flourine is a stronger oxidising agent than chlorine because:

(a) F-F bond has a low enthalpy of dissociation.

(b) Flouride ion (F^-) has high hydration enthalpy.

(c) Electron gain enthalpy of flourine is less negative than chlorine.

(d) Flourine has a very small size.

Choose the most appropriate answer from the options given:

[NEET Re-2022]

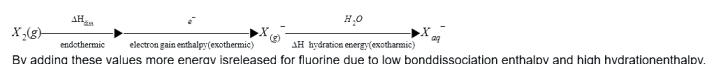
Options:

- A. (b) and (c) only
- B. (a) and (b) only
- C. (a) and (c) only
- D. (a) and (d) only

Answer: B

Solution:

Solution:



Question 8

Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them. [NEET 2021]

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Options:

- A. Noble gases are sparingly soluble in water
- B. Noble gases have very high melting and boiling points
- C. Noble gases have weak dispersion forces
- D. Noble gases have large positive values of electron gain enthalpy

Answer: B

Solution:

Solution:

Noble gases have weak dispersion forces hence they have low melting and boiling points.

Question 9

Statement I : Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.

Statement II : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the correct answer from the options given below.

[NEET 2021]

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Options:

- A. Both statement I and Statement II are true
- B. Both Statement I and Statement II are false
- C. Statement I is correct but statement II is false
- D. Statement I is incorrect but Statement II is true

Answer: A

Solution:

Solution:

In the modern periodic table, moving down the group as the size of halogen atom increases, the H – X bond length also increases as a result the bond enthalpy decreases. Hence, The acidic strength also increases.

So, the correct order of acidic strength is

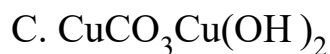
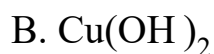
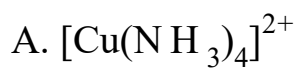
$\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$

Question10

Urea reacts with water to form A which will decompose to form B. B when passed through Cu^{2+} (aq), deep blue colour solution C is formed. What is the formula of C from the following?
(2020)

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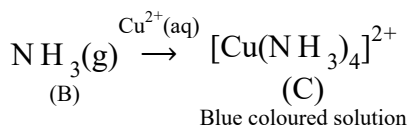
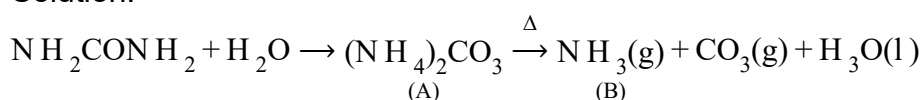
Options:



Answer: A

Solution:

Solution:



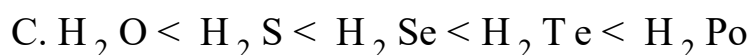
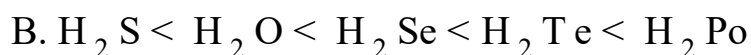
Question11

Which is the correct thermal stability order for H_2E ($\text{E} = \text{O}, \text{S}, \text{Se}, \text{Te}$ and Po)?

(NEET 2019)

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Options:



Answer: D

Solution:

Solution:

The thermal stability of hydrides decreases from H_2O to H_2Po . This is because as the size of atom E in H_2E increases, the bond $\text{H}-\text{E}$ becomes weaker and thus breaks on heating.

Therefore, the correct order of thermal stability is $\text{H}_2\text{Po} < \text{H}_2\text{Te} < \text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{O}$

Question12

Match the Xenon compounds in Column-I with its structure in Column-II and assign the correct code.

Column-I	Column-II
(A) XeF_4	(i) pyramidal
(B) XeF_6	(ii) square planar
(C) XeOF_4	(iii) distorted octahedral
(D) XeO_3	(iv) square pyramidal

(NEET 2019)

Options:

A. (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

B. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

C. (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)

D. (A)-(ii), (B)-(iii), (C)-(i), (D)-(iv)

Answer: C

Solution:

Solution:

Question13

Which of the following oxoacid of sulphur has $-\text{O}-\text{O}-$ linkage?
[2020]

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Options:

A. H_2SO_4 , sulphuric acid

B. $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid

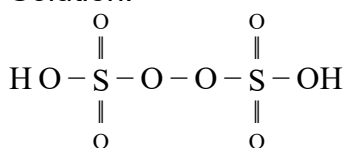
C. $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid

D. H_2SO_3 , sulphurous acid

Answer: B

Solution:

Solution:



Peroxodisulphuric acid

Question14

Match the following:

(A) Pure nitrogen	(i) Chlorine
(B) Haber process	(ii) Sulphuric acid
(C) Contact process	(iii) Ammonia
(D) Deacon's process	(iv) Sodium azide or Barium azide

(NEET 2019)

Options:

A. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)

B. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

C. (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)

D. (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

Answer: A

Solution:

Solution:

Question15

**Identify the incorrect statement related to PCl_5 from the following :
(NEET 2019)**

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Options:

A. PCl_5 molecule is non-reactive.

B. Three equatorial P – Cl bonds make an angle of 120° with each other.

C. Two axial P – Cl bonds make an angle of 180° with each other.

D. Axial P – Cl bonds are longer than equatorial P – Cl bonds.

Answer: A

Solution:

Solution:

Due to longer and hence weaker axial bonds, PCl_5 is a reactive molecule.

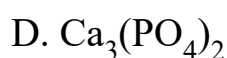
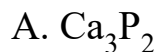
Question16

A compound 'X' upon reaction with H_2O produces a colourless gas 'Y' with rotten fish smell. Gas 'Y' is absorbed in a solution of CuSO_4

to give Cu_3P_2 as one of the products. Predict the compound 'X'.
(Odisha NEET 2019)

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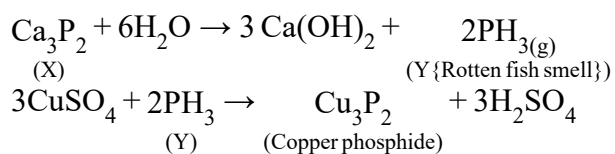
Options:



Answer: A

Solution:

Solution:

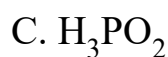
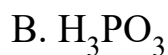


Question17

Which of the following oxoacids of phosphorus has strongest reducing property?
(Odisha NEET 2019)

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Options:



Answer: C

Solution:

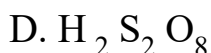
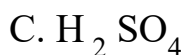
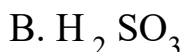
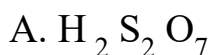
Solution:

Acids which contain P – H bonds have strong reducing properties. Among the given compounds, H_3PO_2 is the strongest reducing agent as it contains two P – H bonds.

Question18

**Identify the correct formula of oleum from the following :
(Odisha NEET 2019)**

Options:



Answer: A

Solution:

Solution:

Question19

**Which of the following statements is not true for halogens?
(NEET 2018)**

Options:

A. All form monobasic oxyacids.

- B. All are oxidizing agents.
- C. All but fluorine show positive oxidation states.
- D. Chlorine has the highest electron-gain enthalpy.

Answer: C

Solution:

Solution:

All halogens show both positive and negative oxidation states while fluorine shows only negative oxidation state except +1 in HOF.

Question20

Match the interhalogen compounds of column-I with the geometry in column-II and assign the correct code.

Column I	Column II
(A) XX'	(i) T-shape
(B) XX'_3	(ii) Pentagonal bipyramidal
(C) XX'_5	(iii) Linear
(D) XX'_7	(iv) Square pyramidal
	(v) Tetrahedral

(NEET 2017)

Options:

- A. A-(iii), B-(i), C-(iv), D-(ii)
- B. A-(v), B-(iv), C-(iii), D-(ii)
- C. A-(iv), B-(iii), C-(ii), D-(i)
- D. A-(iii), B-(iv), C-(i), D-(ii)

Answer: A

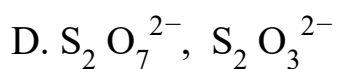
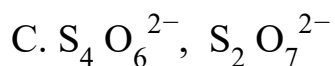
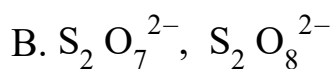
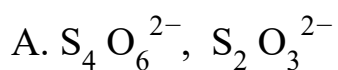
Solution:

Solution:

Question21

**In which pair of ions both the species contain S – S bond?
(NEET 2017)**

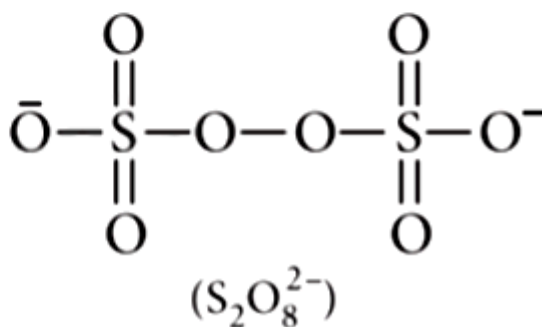
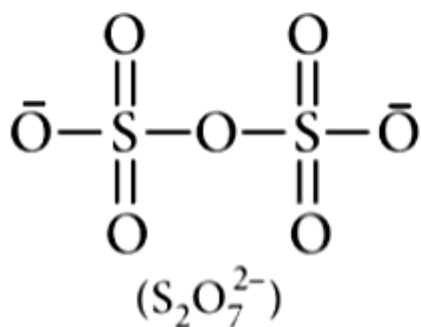
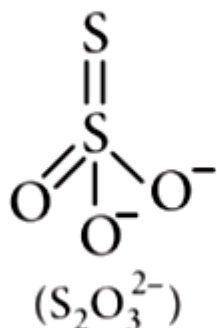
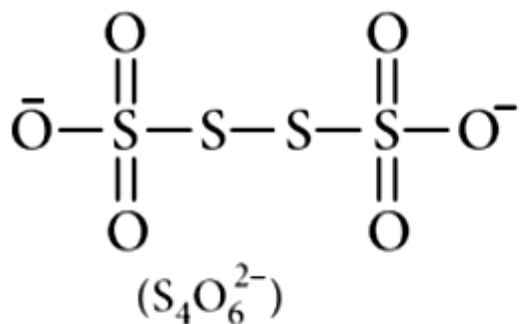
Options:



Answer: A

Solution:

Solution:



Question22

Match the compounds given in column I with the hybridisation and shape given in column II and mark the correct option.

Column I

Column II

- | | |
|---------------------|--------------------------|
| (A) XeF_6 | (i) distorted octahedral |
| (B) XeO_3 | (ii) square planar |
| (C) XeOF_4 | (iii) pyramidal |
| (D) XeF_4 | (iv) square pyramidal |

(NEET-I 2016)

Options:

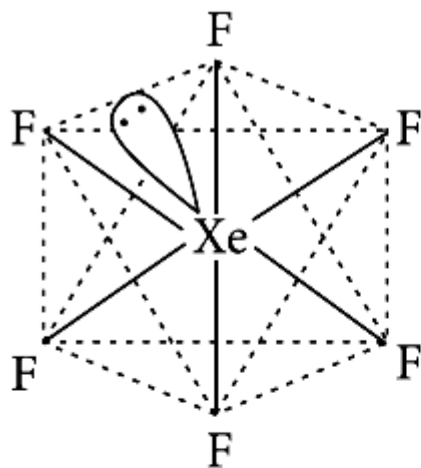
- A. A-(iv), B-(iii), C-(i), D-(ii)
- B. A-(iv), B-(i), C-(ii), D-(iii)
- C. A-(i), B-(iii), C-(iv), D-(ii)

D. A-(i), B-(ii), C-(iv), D-(iii)

Answer: C

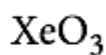
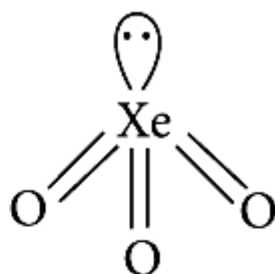
Solution:

Solution:



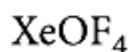
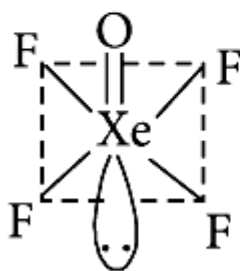
Distorted octahedral

sp^3d^3



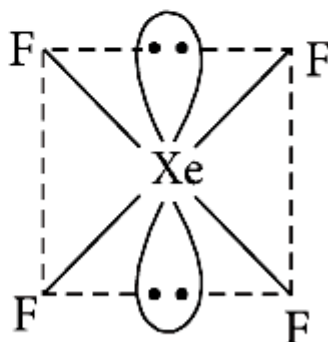
Pyramidal

sp^3



Square pyramidal

sp^3d^2



Square planar

sp^3d^2

Question23

**Which is the correct statement for the given acids?
(NEET-I 2016)**

Options:

- A. Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid.
- B. Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid.

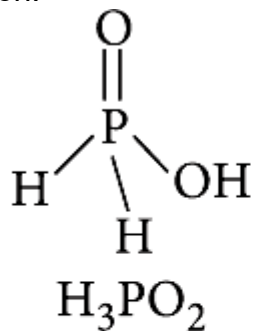
C. Both are diprotic acids.

D. Both are triprotic acids.

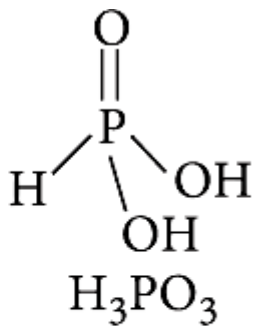
Answer: A

Solution:

Solution:



Hypophosphorous acid
or Phosphinic acid
(Monobasic)



Orthophosphorous acid
or Phosphonic acid
(Dibasic)

Question24

Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules?
(NEET-I 2016)

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Options:

A. $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$

B. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$

C. $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$

D. $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$

Answer: D

Solution:

Solution:

The order of bond dissociation enthalpy is : $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$

Bond dissociation energy of halogen family decreases down the group as the size of atom increases.

The decreasing order for the bond dissociation enthalpy of halogen molecules is

$\text{Cl}_2 57 \text{ kcal/mol} > \text{Br}_2 45.5 \text{ kcal/mol} > \text{F}_2 38 \text{ kcal/mol} > 35.6 \text{ kcal/mol I}_2$.

A halogen molecule having larger atoms should have low dissociation energy and vice versa. Fluorine is an exception because of interelectronic repulsion is present in small atom fluorine.

Question 25

When copper is heated with conc. HNO_3 it produces (NEET-I 2016)

Options:

A. $\text{Cu}(\text{NO}_3)_2$, NO and NO_2

B. $\text{Cu}(\text{NO}_3)_2$ and N_2O

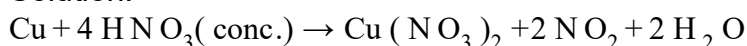
C. $\text{Cu}(\text{NO}_3)_2$ and NO_2

D. $\text{Cu}(\text{NO}_3)_2$ and NO

Answer: C

Solution:

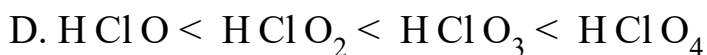
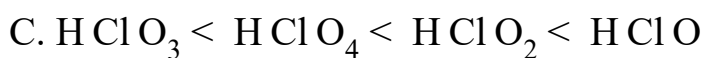
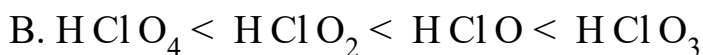
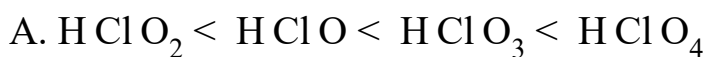
Solution:



Question 26

Among the following, the correct order of acidity is (NEET-I 2016, 2007, 2005)

Options:



Answer: D

Solution:

Solution:

15. (d) : The acidic character of the oxoacids increases with increase in oxidation number of the halogen atom i.e., $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$.

This can be explained on the basis of relative stability of the anions left after removal of a proton. since the stability of the anion decreases in the order : $\text{ClO}_4^- > \text{ClO}_3^- > \text{ClO}_2^- > \text{ClO}^-$, acid strength also decreases in the same order.

Question27

Strong reducing behaviour of H_3PO_2 is due to (2015)

Options:

A. high electron gain enthalpy of phosphorus

B. high oxidation state of phosphorus

C. presence of two -OH groups and one P—H bond

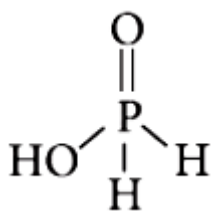
D. presence of one -OH group and two P—H bonds.

Answer: D

Solution:

Solution:

All oxyacids of phosphorus which have P—H bonds act as strong reducing agents. H_3PO_2 has two P—H bonds hence, it acts as a strong reducing agent.



Question28

The variation of the boiling points of the hydrogen halides is in the order $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$.

What explains the higher boiling point of hydrogen fluoride?
(2015)

©

Options:

- A. There is strong hydrogen bonding between HF molecules.
- B. The bond energy of HF molecules is greater than in other hydrogen halides
- C. The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule
- D. The electronegativity of fluorine is much higher than for other elements in the group.

Answer: A

Solution:

Solution:

HF forms strong intermolecular H-bonding due to high electronegativity of F. Hence, the boiling point of HF is abnormally high. Boiling points of other hydrogen halides gradually increase from HCl to HI due to increase in size of halogen atoms from Cl to I which further increase the magnitude of van der Waals forces.

Question29

Which of the statements given below is incorrect?
(2015)

Options:

- A. O_3 molecule is bent.
- B. ONF is isoelectronic with O_2N^-
- C. OF_2 is an oxide of fluorine
- D. Cl_2O_7 is an anhydride of perchloric acid

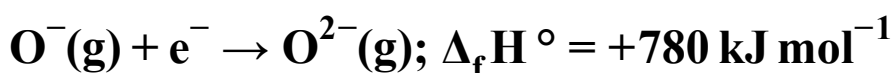
Answer: C**Solution:**

Solution:

OF_2 (oxygen difluoride) is a fluoride of oxygen because fluorine is more electronegative than oxygen.

Question 30

The formation of the oxide ion, $O^{2-}(g)$ from oxygen atom requires first an exothermic and then an endothermic step as shown below :



Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that, (2015)

Options:

- A. O^- ion has comparatively smaller size than oxygen atom
- B. oxygen is more electronegative
- C. addition of electron in oxygen results in larger size of the ion

D. electron repulsion outweighs the stability gained by achieving noble gas configuration.

Answer: D

Solution:

Solution:

Question31

Nitrogen dioxide and sulphur dioxide have some properties in common. Which property is shown by one of these compounds, but not by the other?
(2015 Cancelled)

©

Options:

- A. Is soluble in water.
- B. Is used as a food preservative
- C. Forms 'acid-rain'
- D. Is a reducing agent.

Answer: B

Solution:

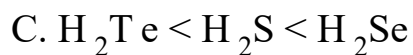
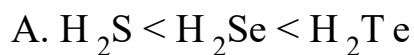
Solution:
 NO_2 is not used as a food preservative.

Question32

Acidity of diprotic acids in aqueous solutions increases in the order
(2014)

©

Options:



Answer: A

Solution:

Solution:

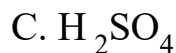
As the atomic size increases down the group, the bond length increases and the bond strength decreases and the cleavage of E-H bond becomes easier thus, more will be the acidity. Thus, the correct order is: $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$

Question33

**Which is the strongest acid in the following?
(2013 NEET)**

©

Options:



Answer: A

Solution:

Solution:

$\text{H}\overset{7}{\text{Cl}}\text{O}_4$ with highest oxidation number and its conjugate base is resonance stabilised, hence it is most acidic. Cl is more electronegative than S.

Question34

Which one of the following molecules contains no π bond?
(2013 NEET)

Options:

A. SO_2

B. NO_2

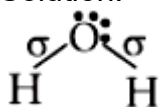
C. CO_2

D. H_2O

Answer: D

Solution:

Solution:



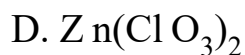
Question35

Which of the following does not give oxygen on heating?
(NEET 2013)

Options:

A. $\text{K}_2\text{Cr}_2\text{O}_7$

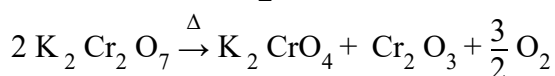
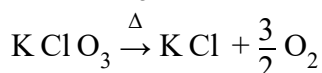
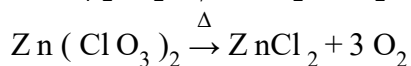
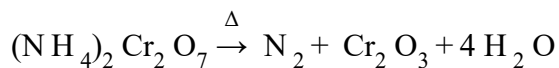
B. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$



Answer: B

Solution:

Solution:



Question36

**Which of the following statements about the interstitial compounds is incorrect?
(2013 NEET)**

©

Options:

- A. They are much harder than the pure metal.
- B. They have higher melting points than the pure metal.
- C. They retain metallic conductivity.
- D. They are chemically reactive.

Answer: D

Solution:

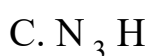
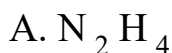
Solution:

Interstitial compounds are generally chemically inert

Question37

In which of the following compounds, nitrogen exhibits highest oxidation state?
(NEET 2013)

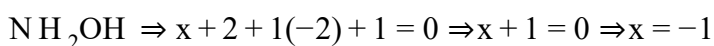
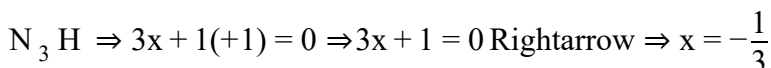
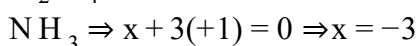
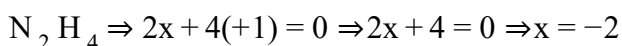
Options:



Answer: C

Solution:

Solution:



Thus, highest oxidation state is $-\frac{1}{3}$.

Question38

Which of the following statements is not valid for oxoacids of phosphorus?
(2012)

Options:

A. Orthophosphoric acid is used in the manufacture of triple superphosphate

B. Hypophosphorous acid is a diprotic acid

C. All oxoacids contain tetrahedral four coordinated phosphorus.

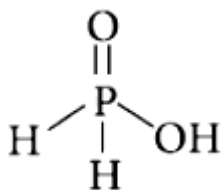
D. All oxoacids contain atleast one P = O unit and one P-OH group

Answer: B

Solution:

Solution:

Hypophosphorous acid is a monoprotic acid.

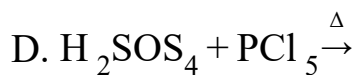
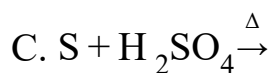
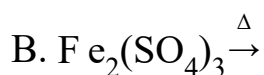


Question39

Sulphur trioxide can be obtained by which of the following reaction? (2012)

©

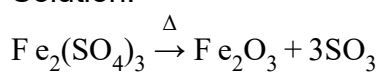
Options:



Answer: B

Solution:

Solution:



Question40

**In which of the following arrangements the given sequence is not strictly according to the property indicated against it?
(2012 Mains)**

Options:

- A. $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: increasing acidic strength
- B. $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: increasing pK_a values
- C. $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: increasing acidic character
- D. $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: increasing oxidising power

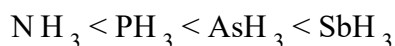
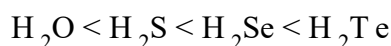
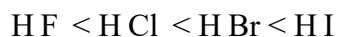
Answer: B

Solution:

Solution:

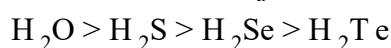
Acidic strength of hydrides increase with increase in molecular mass.

Thus order of acidic strength is



And as acidic strength increases pK_a decreases.

Thus order of pK_a



Question41

**Oxidation states of P in $\text{H}_4\text{P}_2\text{O}_5$, $\text{H}_4\text{P}_2\text{O}_6$, $\text{H}_4\text{P}_2\text{O}_7$ are respectively
(2010)**

Options:

- A. +3, +5, +4
- B. +5, +3, +4

C. +5, +4, +3

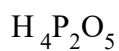
D. +3, +4, +5

Answer: D

Solution:

Solution:

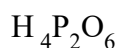
The oxidation state can be calculated as :



$$+4 + 2x + 5(-2) = 0$$

$$2x - 6 = 0$$

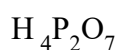
$$x = +3$$



$$+4 + 2x + 6(-2) = 0$$

$$2x - 8 = 0$$

$$x = +4$$



$$+4 + 2x + 7(-2) = 0$$

$$2x - 10 = 0$$

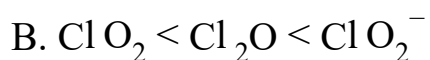
$$2x = 10$$

$$x = +5$$

Question42

**The correct order of increasing bond angles in the following species is
(2010)**

Options:



Answer: D

Solution:

Solution:

Question43

How many bridging oxygen atoms are present in P_4O_{10} (2010 Mains)

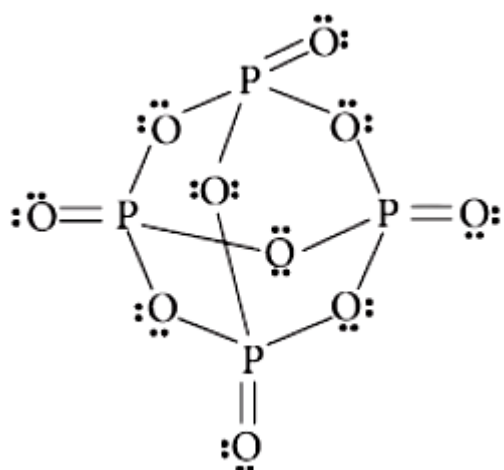
Options:

- A. 6
- B. 4
- C. 2
- D. 3

Answer: A

Solution:

Solution:



Question44

Among the following which is the strongest oxidising agent? (2009)

Options:

A. Br_2

B. I_2

C. Cl_2

D. F_2

Answer: D

Solution:

Solution:

Standard reduction potentials of halogens are positive and decrease from fluorine to iodine. So, F_2 is the strongest oxidising agent

Question 45

The angular shape of ozone molecule (O_3) consists of (2008)

Options:

A. 1 σ and 1 π bond

B. 2 σ and 1 π bond

C. 1 σ and 2 π bonds

D. 2 σ and 2 π bonds

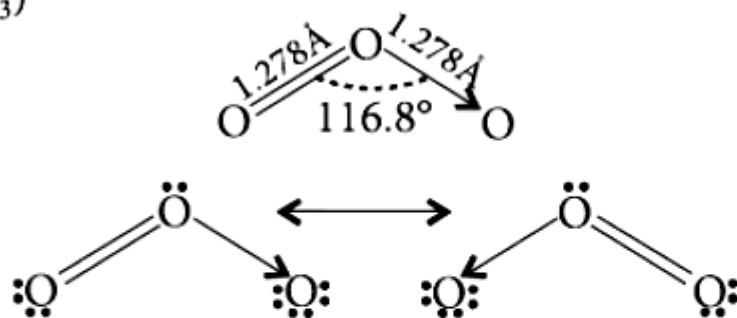
Answer: B

Solution:

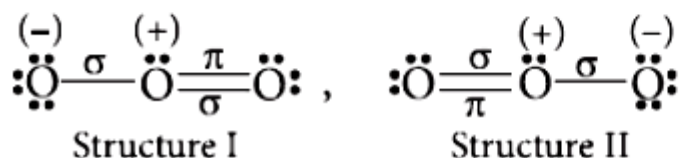
Solution:

The angular shape of ozone molecule (O_3)

13)



O_3 molecules can be represented by the following two Lewis structures.



Question46

The electronegativity difference between N and F is greater than that between N and H yet the dipole moment of NH_3 (1.5D) is larger than that of NF_3 (0.2D). This is because (2006)

Options:

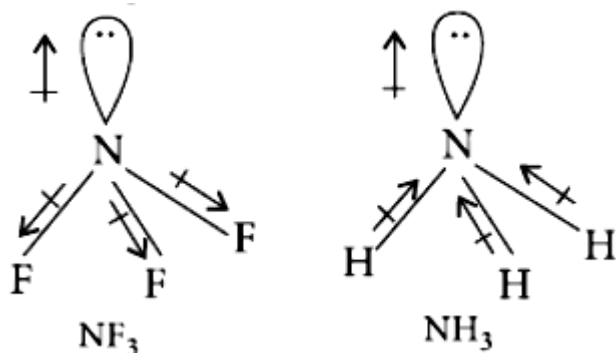
- A. in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction
- B. in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction
- C. in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions
- D. in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in opposite directions.

Answer: C

Solution:

Solution:

The dipole moment of NF_3 0.24 D and of NH_3 is 1.48 D. The difference is due to fact that while the dipole moment due to N - F bonds in NF_3 are in opposite direction to the direction of the dipole moment of the lone pair on N atom which partly cancel out, the dipole moment of N - H bonds in NH_3 are in the same direction of the dipole moment of the lone pair on N atom which adds up as shown below.



Question47

Which one of the following orders is not in accordance with the property stated against it?

(2006)

Options:

- A. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Bond dissociation energy
- B. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Oxidising power
- C. $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$: Acidic property in water
- D. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$: Electronegativity

Answer: A

Solution:

Solution:

X - X bond	F - F	Cl - Cl	Br-Br	I-I
Bond dissociation energy(kcl/mol)	38	57	45.5	35.6

The lower value of bond dissociation energy of fluorine is due to the high inter-electronic repulsions between non-bonding electrons in the 2p-orbitals of fluorine. As a result F - F bond is weaker in comparison to Cl Cl and Br - Br bonds.

Question48

In which of the following molecules are all the bonds are not equal?
(2006)

©

Options:



Answer: B

Solution:

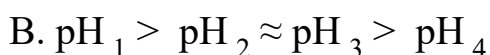
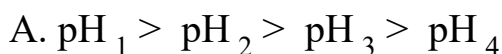
Solution:

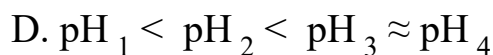
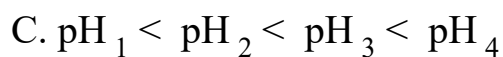
The $\text{Cl}-\text{F}$ ($\text{Cl}-\text{F}_b$) bond length is equal to $1.60\text{\AA}$ while each of the two axial $\text{Cl}-\text{F}$ ($\text{Cl}-\text{F}_a$) bond length is equal to $1.70\text{\AA}$.

Question49

What is the correct relationship between the pH of isomolar solutions of sodium oxide, Na_2O (pH_1), sodium sulphide, Na_2S (pH_2), sodium selenide, Na_2Se (pH_3) and sodium telluride Na_2Te (pH_4)?
(2005)

Options:

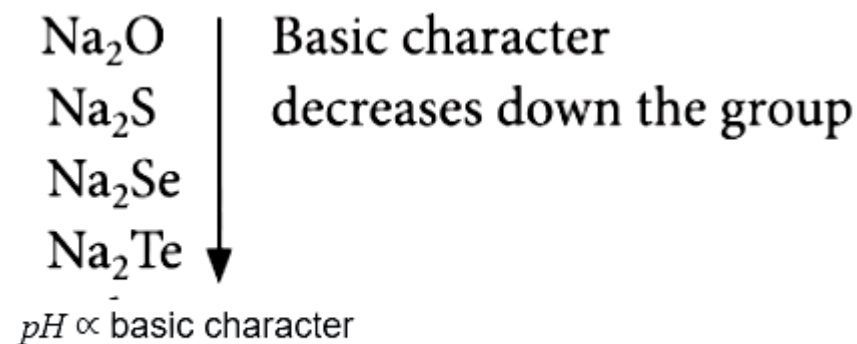




Answer: A

Solution:

Solution:

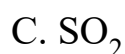
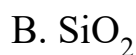
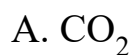


Hence, $\text{pH}_1 > \text{pH}_2 > \text{pH}_3 > \text{pH}_4$

Question 50

**Which one of the following oxides is expected to exhibit paramagnetic behaviour?
(2005)**

Options:



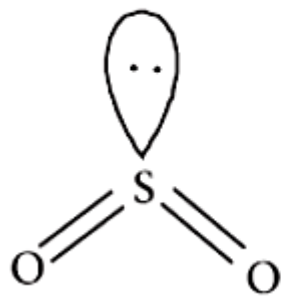
Answer: D

Solution:

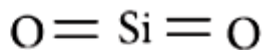
Solution:



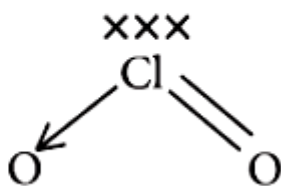
Diamagnetic



Diamagnetic



Diamagnetic

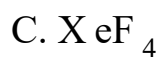
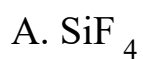


Paramagnetic

Question51

Which of the following would have a permanent dipole moment?
(2005)

Options:

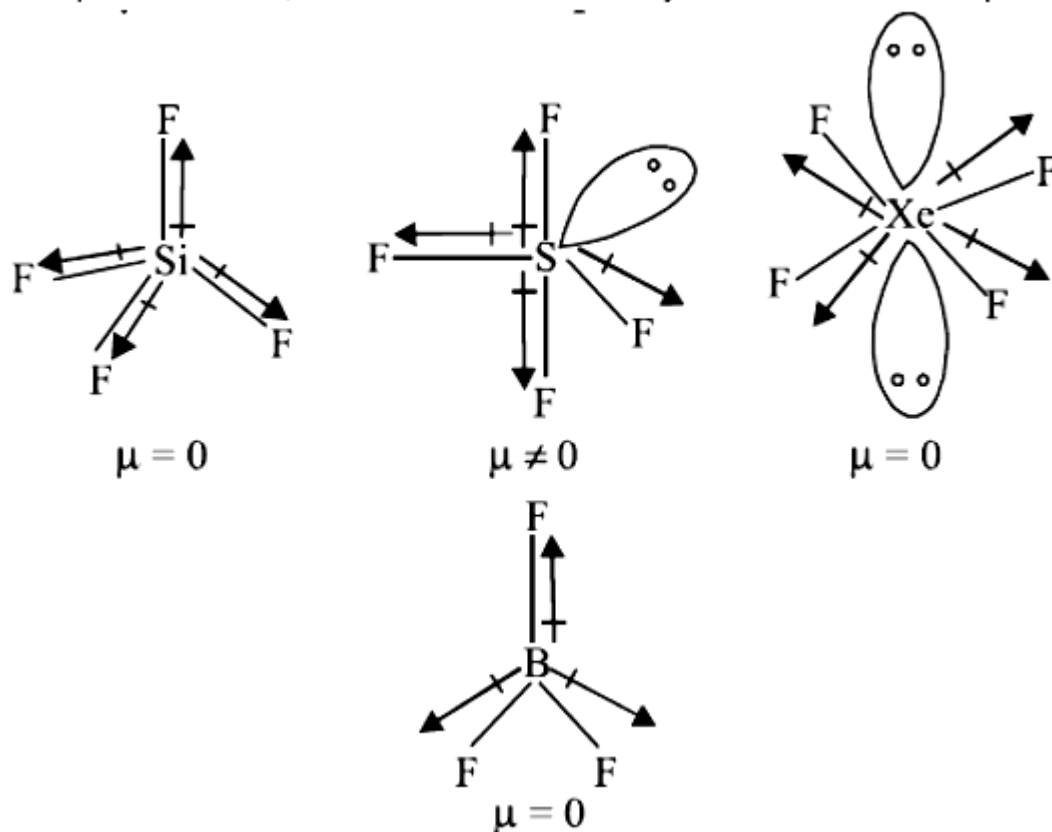


Answer: B

Solution:

Solution:

For dipole moment, we have to know the hybridisation and shape.



Question52

Which of the following statements is true?
(2002)

Options:

- A. Silicon exhibits 4 coordination number in its compound.
- B. Bond energy of F₂ is less than Cl₂.
- C. Mn(III) oxidation state is more stable than Mn(II) in aqueous state.
- D. Elements of 15th gp shows only +3 and +5 oxidation states.

Answer: B

Solution:

Solution:

Fluorine is more reactive than chlorine. So bond energy of chlorine is greater than fluorine.

Silicon exhibits coordination number 6.

In aqueous state Mn(II) is more stable.



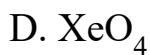
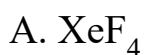
The common oxidation states of 15th group elements are -3,+3 and +5 .

Question53

**Which compound has planar structure?
(2000)**

©

Options:



Answer: A

Solution:

Solution:

In XeF_4 the Xe atom is sp^3d^2 hybridised, which contains two lone pair orbitals and four bond pair orbitals. Therefore the shape of XeF_4 molecule is square planar, with one lone pair orbital over and other below the plane.

Question54

**Which of the following oxides is most acidic?
(1999)**

©

Options:

A. As_2O_5

B. P_2O_5

C. N_2O_5

D. Sb_2O_5

Answer: C

Solution:

Solution:

As among N, P, As and Sb, the former has highest electronegativity (EN) so its oxide is most acidic.

As the electronegativity value of element increases, the acidic character of the oxide also increases.

The oxide with the highest positive oxidation state on the element other than O should be most acidic. Oxidation state of V in V_2O_5 and N in N_2O_5 are same. But the electronegativity of N is higher, making N_2O_5 the most acidic oxide.

Question55

**Which of the following phosphorus is the most reactive?
(1999)**

©

Options:

A. Scarlet phosphorus

B. White phosphorus

C. Red phosphorus

D. Violet phosphorus

Answer: B

Solution:

Solution:

White phosphorus has low ignition temperature so it is most reactive among all the allotropes.

Question56

**Which of the following is used in the preparation of chlorine?
(1999)**

©

Options:

- A. Both MnO_2 and KMnO_4
- B. Only KMnO_4
- C. Only MnO_2
- D. Either MnO_2 or KMnO_4

Answer: A

Solution:

Solution:



Question57

**Repeated use of which one of the following fertilizers would increase the acidity of the soil?
(1998)**

©

Options:

- A. Ammonium sulphate
- B. Superphosphate of lime

C. Urea

D. Potassium nitrate

Answer: A

Solution:

Solution:

Ammonium sulphate is a salt of strong acid (H_2SO_4) and weak base (NH_4OH) Therefore, repeated use of ammonium sulphate would increase the concentration of sulphuric acid, while ammonia from NH_4OH is used up by the plant. Hence, the acidity of soil will increase.

Question58

**Which of the following has the highest dipole moment?
(1997)**

©

Options:

A. SbH_3

B. AsH_3

C. NH_3

D. PH_3

Answer: C

Solution:

Solution:

Due to greater electronegativity of nitrogen, dipole moment for NH_3 is greater.

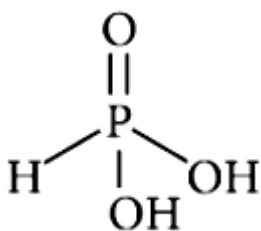
Question59

The structural formula of hypophosphorous acid is

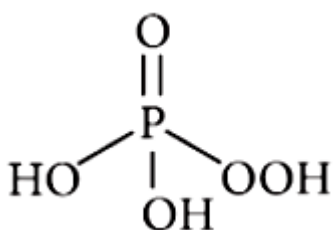
(1997)

Options:

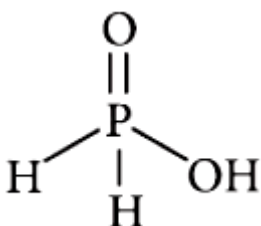
A.



B.



C.



D. None of these.

Answer: C

Solution:

Solution:

The formula of hypophosphorous acid is H_3PO_2 as shown in (c). It is a monobasic acid.

Question60

Which of the following bonds has the highest energy?
(1996)

Options:

A. S-S

B. O-O

C. Se-Se

D. Te-Te

Answer: A

Solution:

Solution:

B . E.	O – O	S – S	Se – Se	Te – Te
(kJ mol ⁻¹):	142	226	172	126

Question61

**The basic character of hydrides of the V group elements decreases in the order
(1996)**

©

Options:

A. $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$

B. $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$

C. $\text{SbH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{NH}_3$

D. $\text{NH}_3 > \text{SbH}_3 > \text{PH}_3 > \text{AsH}_3$

Answer: A

Solution:

Solution:

All the hydrides of group V elements have one lone pair of electrons on their central atom. Therefore, they can act as Lewis bases. The basic character of these hydrides decreases down the

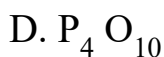
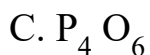
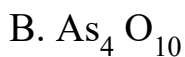
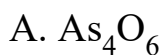
group.

Question62

Among the following oxides, the lowest acidic is (1996)

©

Options:



Answer: A

Solution:

Solution:

The acidic character of the oxides decreases with the decrease in the oxidation state and also decreases down the group.

Question63

Which of the following has the greatest electron affinity? (1996)

©

Options:

A. I

B. Br

C. F

D. Cl

Answer: D

Solution:

Solution:

In general, the electron affinity decreases from top to bottom in a group. But in group 17, fluorine has lower electron affinity as compared to chlorine due to very small size of fluorine atom.

Question64

**Which of the following represents calcium chlorite?
(1996)**

©

Options:

A. $\text{Ca}(\text{ClO}_3)_2$

B. $\text{Ca}(\text{ClO}_2)_2$

C. CaClO_2

D. $\text{Ca}(\text{ClO}_4)_2$

Answer: B

Solution:

Solution:

since the valency of calcium is 2 and a chlorite ion is ClO_2^- , the calcium chlorite is $\text{Ca}(\text{ClO}_2)_2$.

Question65

**Reaction of sodium thiosulphate with iodine gives
(1996)**

©

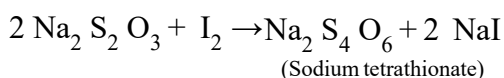
Options:

- A. tetrathionate ion
- B. sulphide ion
- C. sulphate ion
- D. sulphite ion.

Answer: A

Solution:

Solution:



Question66

About 20 km above the earth, there is an ozone layer. Which one of the following statements about ozone and ozone layer is true?
(1995)

©

Options:

- A. It is beneficial to us as it stops U.V. radiation.
- B. Conversion of O_3 to O_2 is an endothermic reaction.
- C. Ozone is a triatomic linear molecule.
- D. It is harmful as it stops useful radiation.

Answer: A

Solution:

Solution:

Ozone layer is very beneficial to us, because it stops harmful ultraviolet radiations to reach the earth.

Question67

The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^3$. What is the atomic number of the element, which is just below the above element in the periodic table?
(1995)

©

Options:

- A. 36
- B. 49
- C. 33
- D. 34

Answer: C

Solution:

Solution:

Atomic number of the given element is 15 and it belongs to 5th group. Therefore atomic number of the element below the above element = $15 + 18 = 33$.

Question68

Which of the following oxides of nitrogen is paramagnetic?
(1994)

©

Options:

- A. NO_2
- B. $\text{N}_2 \text{O}_3$
- C. $\text{N}_2 \text{O}$

D. N_2O_5

Answer: A

Solution:

Solution:

NO_2 is paramagnetic due to the presence of unpaired electrons.

Question69

**Which of the following displaces Br_2 from an aqueous solution containing bromide ions?
(1994)**

Options:

A. I_2

B. I_3^-

C. Cl_2

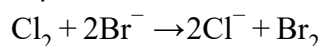
D. Cl^-

Answer: C

Solution:

Solution:

since chlorine is more electronegative than bromine, therefore it will displace bromine from an aqueous solution containing bromide ions.



Question70

Which of the following fluorides does not exist?

(1993)

©

Options:

A. NF_5

B. PF_5

C. AsF_5

D. SbF_5

Answer: A

Solution:

Solution:

Nitrogen cannot form pentahalides because it cannot expand its octet due to non-availability of d - orbitals.

Question71

**Which of the following species has four lone pairs of electrons?
(1993)**

©

Options:

A. I

B. O^-

C. Cl^-

D. He

Answer: C

Solution:

Solution:

Outer electronic configuration of Cl = $3s^2 3p_x^2 3p_y^2 3p_z^1$

Outer electronic configuration of $Cl^- = 3s^2 3p_x^2 3p_y^2 3p_z^2$, i.e., 4 lone pair of electrons

Question72

**Which of the following sets has strongest tendency to form anions?
(1993)**

©

Options:

A. Ga, Ni, Tl

B. Na, Mg, Al

C. N, O, F

D. V, Cr, Mn

Answer: C

Solution:

Solution:

N, O and F are highly electronegative non-metals and will have the strongest tendency to form anions by gaining electrons from metal atoms.

Question73

**A solution of potassium bromide is treated with each of the following. Which one would liberate bromine?
(1993)**

©

Options:

A. Hydrogen iodide

B. Sulphur dioxide

C. Chlorine

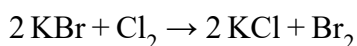
D. Iodine

Answer: C

Solution:

Solution:

A stronger oxidising agent (Cl_2) displaces a weaker oxidising agent (Br_2) from its salt solution.



Question74

**Which of the following elements is extracted commercially by the electrolysis of an aqueous solution of its compound?
(1993)**

©

Options:

A. Cl

B. Br

C. Al

D. Na

Answer: A

Solution:

Solution:

Chlorine is obtained by the electrolysis of brine (concentrated NaCl solution). Chlorine is liberated at anode.

Question75

**Number of electrons shared in the formation of nitrogen molecule is
(1992)**

Options:

- A. 6
- B. 10
- C. 2
- D. 8

Answer: A

Solution:

Solution:

Nitrogen molecule is diatomic containing a triple bond between two N atoms, $\ddot{\text{N}} \equiv \ddot{\text{N}}$ therefore, nitrogen molecule is formed by sharing six electrons.

Question 76

Sugarcane on reaction with nitric acid gives (1992)

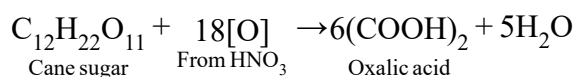
Options:

- A. CO_2 and SO_2
- B. $(\text{COOH})_2$
- C. 2 HCOOH (two moles)
- D. no reaction.

Answer: B

Solution:

Solution:



Question77

Nitrogen is relatively inactive element because (1992)

©

Options:

- A. its atom has a stable electronic configuration
- B. it has low atomic radius
- C. its electronegativity is fairly high
- D. dissociation energy of its molecule is fairly high.

Answer: D

Solution:

Solution:

N_2 molecule contains triple bond between N atoms having very high dissociation energy (946 kJ mol^{-1}) due to which it is relatively inactive.

Question78

H_3PO_2 is the molecular formula of an acid of phosphorus. Its name and basicity respectively are (1992)

Options:

- A. phosphorous acid and two
- B. hypophosphorous acid and two
- C. hypophosphorous acid and one

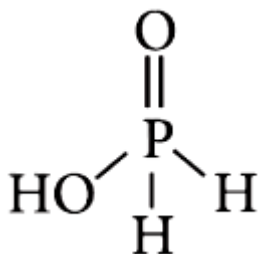
D. hypophosphoric acid and two.

Answer: C

Solution:

Solution:

H_3PO_2 is named as hypophosphorous acid. As it contains only one P – OH group, its basicity is one.



Question79

**Which of the following bonds will be most polar?
(1992)**

©

Options:

A. N-Cl

B. O-F

C. N-F

D. N-N

Answer: C

Solution:

Solution:

Polarity of the bond depends upon the electronegativity difference of the two atoms forming the bond. Greater the electronegativity difference, more is the polarity of the bond.

N – Cl	O – F	N – F	N – N
3.04 – 3.16	3.5 – 4.0	3.04 – 4.0	3.04 – 3.04

Question80

Elements of which of the following groups will form anions most readily?
(1992)

©

Options:

- A. Oxygen family
- B. Nitrogen family
- C. Halogens
- D. Alkali metals

Answer: C

Solution:

Solution:

As halogens have seven electrons (ns^2np^5) in the valence shell, they have a strong tendency to acquire the nearest inert gas configuration by gaining an electron from the metallic atom and form halide ions easily.

Question81

Strongest hydrogen bonding is shown by
(1992)

Options:

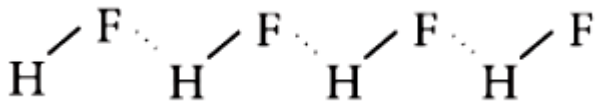
- A. water
- B. ammonia
- C. hydrogen fluoride
- D. hydrogen sulphide.

Answer: C

Solution:

Solution:

Fluorine because of its smaller size and highest electronegativity shows strongest hydrogen bonding.

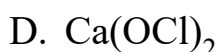
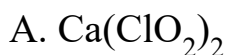


Question82

When chlorine is passed over dry slaked lime at room temperature, the main reaction product is
(1992)

©

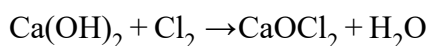
Options:



Answer: C

Solution:

Solution:



Question83

In the manufacture of bromine from sea water, the mother liquor containing bromides is treated with
(1992)

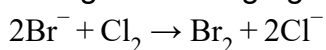
Options:

- A. carbon dioxide
- B. chlorine
- C. iodine
- D. sulphur dioxide.

Answer: B**Solution:**

Solution:

Bromide in the mother liquor (containing MgBr_2) is oxidised to Br_2 by passing Cl_2 which is a stronger oxidising agent.



Question84

**Which would quickly absorb oxygen?
(1991)**

Options:

- A. Alkaline solution of pyrogallol
- B. Conc. H_2SO_4
- C. lime water
- D. alkaline solution of CuSO_4

Answer: A**Solution:**

Solution:

Alkaline solution of pyrogallol absorbs oxygen quickly.

Question85

**Oleum is
(1991)**

Options:

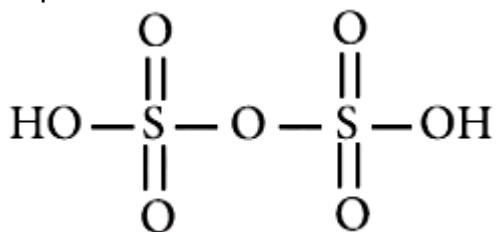
- A. castor oil
- B. oil of vitriol
- C. fuming H_2SO_4
- D. none of these.

Answer: C

Solution:

Solution:

Pyrosulphuric acid or oleum (+6) is $\text{H}_2\text{S}_2\text{O}_7$ which is obtained by dissolving SO_3 and is called fuming sulphuric acid.



Question86

**Aqueous solution of ammonia consists of
(1991)**

Options:

- A. H^+
- B. OH^-

C. NH_4^+

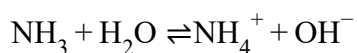
D. NH_4^+ and OH^- .

Answer: D

Solution:

Solution:

Aqueous solution of ammonia contains NH_4^+ and OH^- ions.



Question87

**P_2O_5 is heated with water to give
(1991)**

©

Options:

A. hypophosphorous acid

B. phosphorous acid

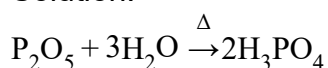
C. hypophosphoric acid

D. orthophosphoric acid.

Answer: D

Solution:

Solution:



Question88

Basicity of orthophosphoric acid is

(1991)

Options:

A. 2

B. 3

C. 4

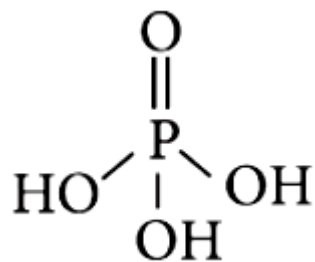
D. 5

Answer: B

Solution:

Solution:

Orthophosphoric acid, H_3PO_4 contains three P – OH groups and is therefore, tribasic.



Question89

PCl_3 reacts with water to form

(1991)

Options:

A. PH_3

B. H_3PO_3 , HCl

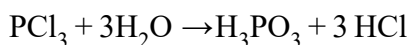
C. POCl_3

D. H_3PO_4

Answer: B

Solution:

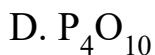
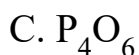
Solution:



Question90

**$\text{PH}_4\text{I} + \text{NaOH}$ forms
(1991)**

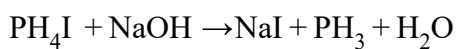
Options:



Answer: A

Solution:

Solution:

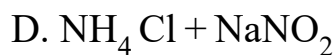
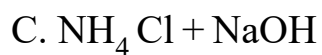


Question91

**Pure nitrogen is prepared in the laboratory by heating a mixture of
(1991)**

Options:

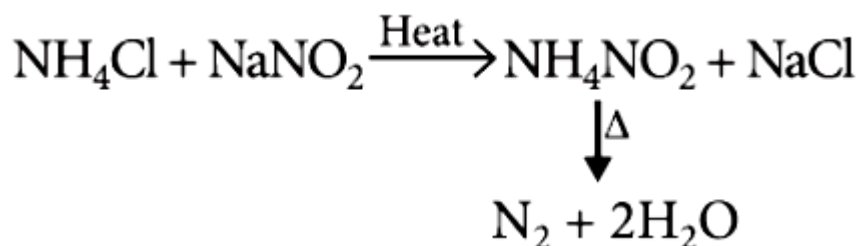




Answer: D

Solution:

Solution:



Question92

The bleaching action of chlorine is due to (1991)

©

Options:

A. reduction

B. hydrogenation

C. chlorination

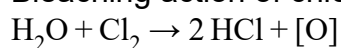
D. oxidation

Answer: D

Solution:

Solution:

Bleaching action of chlorine is due to oxidation in presence of moisture. It is permanent.



Colouring matter + $[\text{O}] \rightarrow$ colourless matter

Question93

Which of the following statement is not correct for nitrogen?
(1990)

©

Options:

- A. Its electronegativity is very high.
- B. d -orbitals are available for bonding.
- C. It is a typical non-metal.
- D. Its molecular size is small.

Answer: B

Solution:

Solution:

In case of nitrogen, d -orbitals are not available for bonding.

N : $1s^2 2s^2 2p^3$

Question94

Which of the following compound does not exist?
(1989)

©

Options:

- A. NCl_5
- B. AsF_5
- C. SbCl_5
- D. PF_5

Answer: A

Solution:

Solution:

All the elements of group 15 form trihalides and pentahalides of the type MX_3 and MX_5 except nitrogen which forms only trihalides. Moreover, nitrogen does not form pentahalides due to the absence of d-orbitals in its valence shell.

Question95

Each of the following is true for white and red phosphorus except that they
(1989)

©

Options:

- A. are both soluble in CS_2
- B. can be oxidised by heating in air
- C. consist of the same kind of atoms
- D. can be converted into one another.

Answer: A

Solution:

Solution:

Red phosphorus is insoluble in CS_2 and only white P is soluble in CS_2 .

Question96

When orthophosphoric acid is heated to 600°C , the product formed is
(1989)

©

Options:

A. PH_3

B. P_2O_5

C. H_3PO_3

D. HPO_3

Answer: D

Solution:

Solution:

On heating, it gives pyrophosphoric acid at 525K and metaphosphoric acid at 875K.

Question97

**Which one has the lowest boiling point?
(1989)**

©

Options:

A. NH_3

B. PH_3

C. AsH_3

D. SbH_3

Answer: B

Solution:

Solution:

Boiling point of hydrides increases with increase in atomic number but ammonia has exceptionally high boiling point due to hydrogen bonding. Thus the correct order of boiling point is,

$\text{BiH}_3 > \text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$

Question98

Oxygen will directly react with each of the following elements except (1989)

©

Options:

- A. P
- B. Cl
- C. Na
- D. S

Answer: B

Solution:

Solution:
Chlorine does not react directly with oxygen.

Question99

The gases respectively absorbed by alkaline pyrogallol and oil of cinnamon are (1989)

©

Options:

- A. O_3 , CH_4
- B. O_2 , O_3
- C. SO_2 , CH_4
- D. N_2O , O_3

Answer: B

Solution:

Solution:

Alkaline pyrogallol absorbs O_2 and oil of cinnamon absorbs O_3 .

Question100

It is possible to obtain oxygen from air by fractional distillation because (1989)

©

Options:

- A. oxygen is in a different group of the periodic table from nitrogen
- B. oxygen is more reactive than nitrogen
- C. oxygen has higher b.pt. than nitrogen
- D. oxygen has a lower density than nitrogen.

Answer: C

Solution:

Solution:

Air is liquefied by making use of the Joule - Thomson effect (cooling by expansion of the gas). Water vapour and CO_2 are removed by solidification. The remaining constituents of liquid air i. e., liquid oxygen and liquid nitrogen are separated by means of fractional distillation (b.pt. of $O_2 = -183^\circ C$; b.pt. of $N_2 = -195.8^\circ C$)

Question101

Bleaching powder reacts with a few drops of conc. HCl to give (1989)

©

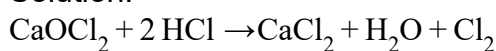
Options:

- A. chlorine
- B. hypochlorous acid
- C. calcium oxide
- D. oxygen.

Answer: A

Solution:

Solution:



Question102

**Which of the following is a nitric acid anhydride?
(1988)**

Options:

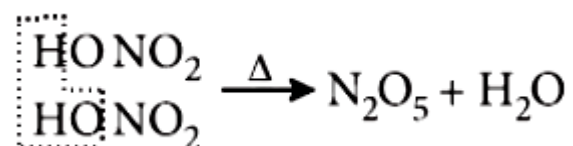
- A. NO
- B. NO₂
- C. N₂O₅
- D. N₂O₃

Answer: C

Solution:

Solution:

When 2 -molecules of nitric acid undergoes heating, loose a water molecule to form an anhydride.



Thus, N₂O₅ is nitric acid anhydride.